ends of an arm supported by a very fine fiber, called a torsion fiber. By measuring how much the fiber gets twisted, one can measure the strength of the force, verify that it is inversely proportional to the square of the distance, and determine how strong it is. Thus, one may accurately determine the coefficient *G* in the formula



Figure 5-10 A cluster of galaxies.

$$F = G\frac{mm'}{r^2}.$$